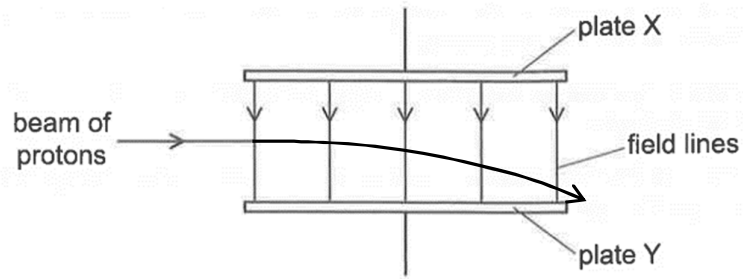


19	D	The proton is positively charged. Plate X is at a higher potential than plate Y. Hence, as the protons enter the uniform electric field between plates X

Qn	Answer	Solution
		and Y, they will experience a constant vertically downward force, hence tracing a parabolic path (non-circular curved path) towards plate Y.  The diagram illustrates a beam of protons entering a region between two parallel horizontal plates, labeled 'plate X' (top) and 'plate Y' (bottom). A horizontal arrow labeled 'beam of protons' enters from the left. Between the plates, five vertical arrows point downwards, labeled 'field lines', representing a uniform electric field. The path of the proton beam is shown as a curved line that starts horizontally and bends downwards towards plate Y as it moves to the right. The entire diagram is enclosed in a rectangular frame.
20	B	Distance SR